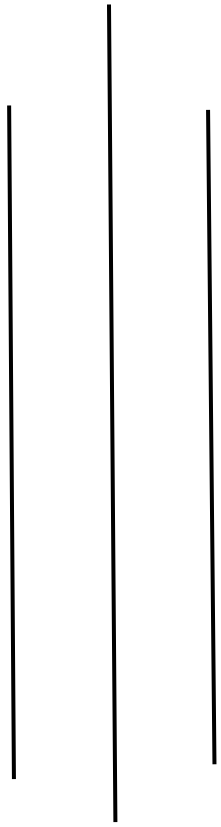
WEEK 2: INTERNSHIP REPORT



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To

OneCover Nepal

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CHAPTER 1: RECONNAISSANCE, SCANNING & ENUMERATION

1.1 Introduction and Objective

Reconnaissance, scanning, and enumeration are important phases of cybersecurity. Reconnaissance involves gathering information about a target system, while scanning is used to identify reachable hosts, open ports, and available services. Enumeration collects detailed information about the services and systems identified during scanning.

This week's practical work focuses on developing basic skills in network reconnaissance, Nmap scanning, service identification, version detection, and safe service enumeration.

- **Objectives**
  - To understand reconnaissance, scanning, and enumeration concepts.
  - To perform host and port discovery using Nmap.
  - To identify running services and their versions.
  - To perform safe service enumeration.
  - To analyze and document scanning and enumeration results.
  - To conduct basic vulnerability research within an authorized lab environment.

1.2 Reconnaissance

Reconnaissance refers to gathering of information about a target system, network or organization to identify potential vulnerabilities and weak points.

Features of Reconnaissance

- Maps digital footprint such as domains, IP addresses and technologies
- It can be passive (stealthy) or active (direct interaction)
- Forms the basis for identifying risks and planning further security actions.

1.2.1 Types of Reconnaissance

a. Passive Reconnaissance

Passive reconnaissance involves gathering information without directly interacting with the target system. The goal is to collect data from publicly available sources to avoid alerting the target to the fact that they are being observed.

Common passive reconnaissance techniques are:

- Reviewing publicly available DNS records.
- Analyzing WHOIS information for domain details.
- Gathering data from social media, public forums, and company websites.
- Monitoring open-source network traffic (e.g. from public Wi-Fi or internet services).

b. Active Reconnaissance

Active reconnaissance involves directly interacting with the target's systems to gather information. This approach is more aggressive and often includes sending probes or testing system responses.

Common active reconnaissance techniques are:

- **Port Scanning**: using tools like Nmap to find open port and services.
- **Ping Sweeps:** To discover live hosts on a network.
- **Banner Grabbing:** To identify software versions or services running on a machine.
- **Vulnerability Scanning:** To detect security weakness.

Reconnaissance Technique and Tools

i. Google Dorking

Google Dorking also called Google Hacking used advances search operators on Google to find hidden or sensitive information that is accidentally exposed online.

Common Search Operators are:

- **site:** search within a specific domain
- **filetype:** searches for a specific file extension like: pdf, txt, xls sql.
- **intitle:** looks for specific word in the title of a web page
- **inurl:** finds specific text stings inside a URL path
- **intext:** looks for specific words in the main body text of a page

ii. Shodan

Shodan is a search engine for internet-connected devices and services**.** It collects information about systems that are reachable on the Internet. It finds: IP address, Open ports, Running Services, Operating System information, Geographical information, Server banners, Service/Software versions.

iii. Footprinting and Fingerprinting

Footprinting is the process of collecting as much information as possible about a target before deeper security testing. During footprinting mainly, we look for:

- Domain names, subdomains, IP addresses, DNS information, Technologies used, Network used, publicly exposed services, Network ranges.

Fingerprinting is more specific. It attempts to identify the exact technology, operating system, service, or version running on a target. During fingerprinting we look for:

- Specific technology identification, identifies system/sevices, os, software versions.

iv. WHOIS lookup

WHOIS lookup is a reconnaissance information-gathering technique used to obtain registration information about a domain name or IP address**.** Depending on the domain, registrar, and privacy protections, a WHOIS lookup may show: Domain name, Registrar, Registration date, Expiration date, Domain status, Nameservers, Registrant organization.

1.3 Scanning

Scanning in cybersecurity is the active process of probing networks, systems, or applications to discover live hosts, open ports, running services, and potential security weaknesses. It acts as a critical bridge between passive reconnaissance and deep vulnerability management.

Major goals of Scanning are:

i. Host Discovery
ii. Open Ports
iii. Service/Protocols
iv. Software + Versions
v. OS Banner Grabbing

Types of Scanning are:

i. **Network Scanning:** It identifies active devices and systems within a network.
ii. **Port Scanning:** It checks which network ports are accessible on a target system.
iii. **Vulnerability Scanning:** It checks for the security weaknesses like: outdated software, missing patches, weak configuration, insecure protocols, exposed devices.
iv. **Network Vulnerability Scanning:** It focuses on network devices and it can examine: routers, switches, firewalls, servers, network appliances.
v. **Web Application Scanning:** It examines websites and web applications for potential security problems, common areas examined are: authentication, session management, input validation, security headers, TLS configuration, Access controls.
vi. **Wireless Network Scanning:** It examines Wi-Fi environments and can identify information such as: nearby wireless networks, SSIDs, Channels, Encryption types, Access points, Signal information.

1.4 Enumeration

Enumeration is the active process of connecting directly to a target system or network to gather detailed information like usernames, open ports, software versions, and shared resources.

Some of the types of Enumeration are:

Table 1.1: Enumeration Types

S.N.	Types	Purpose
1	NetBIOS (Network Basic Input/Output System) Enumeration	Finds Windows computer names, users, domains, and shared resources.
2	SNMP (Simple Network Management Protocol) Enumeration	Collects information about network devices, interfaces, routing, and topology.
3	LDAP (Lightweight Directory Access Protocol) Enumeration	Finds users, groups, email addresses, and directory information.
4	NTP (Network Time Protocol) Enumeration	Discovers time servers, connected hosts, and network information.
5	SMTP (Simple Network Management Protocol) Enumeration	Identifies valid email accounts and mail server details.
6	DNS (Domain Name System) Enumeration	Discovers domains, subdomains, IP addresses, and DNS records.
7	IPsec (Internet Protocol Security) Enumeration	Identifies VPN gateways and their security/configuration details.
8	VoIP (Voice over IP) Enumeration	Discovers SIP servers, extensions, gateways, and VoIP devices.
9	RPC (Remote Procedure Call) Enumeration	Finds running services, endpoints, and accessible applications.
10	Unix/Linux User Enumeration	Identifies user accounts, logged-in users, and system information.
11	SMB (Server Message Block) Enumeration	Discovers shared folders, users, permissions, and Windows/Samba resources.

1.5 Nmap

Nmap (Network Mapper) is an open-source tool used for network discovery, scanning and security auditing. It helps identify hosts, open ports, running services, operating systems and potential vulnerabilities, making it widely used in penetration testing and network troubleshooting.

Features of Nmap are:

- Comprehensive Scanning
- Scripting Engine
- OS Detection
- Service and Version Detection
- Output Formats

Importance of Nmap:

- Security Assessment
- Intrusion Detection
- Network Troubleshooting
- Vulnerability Scanning

Port Range syntax in Nmap

Table 1.2: Port Range Syntax in Nmap

Syntax	Example	Meaning
Single	-p 80	Port 80 only
Multiple	-p 80, 443, 8080	Specific port
Range	-p 1-1024	Ports between 1-1024
All	-p-	All 65,535 ports
Top N	--top-ports 100	Most common ports

Essential Nmap flags

Table 1.3: Flags in Nmap

Flag	Purpose
-sS	TCP SYN Scan
-sT	TCP Connect Scan
-sU	UDP scan
-sn	Host discovery only
-Pn	Skip host discovery
-F	Fast scan
-sv	Detect service/version
-o	OS detection
-A	Aggressive detection
-sC	Default NSE scripts

1.6 Scope and Authorization

This practical was limited to the target machine provided in the TryHackMe Network Services room. The testing was performed only on the authorized lab with no testing of unrelated or public systems.

CHAPTER 2: Lab Environment and Methodology

2.1 Required Lab

The practical was performed in the TryHackMe Network Services room. A target machine was deployed from the room and used as the authorized target for the practical activities.

2.2 Target Environment

The target machine provided by TryHackMe was used for the practical. The assigned target IP was used for connectivity testing, Nmap scanning, service detection, and enumeration.

Table 2.1: Target Environment Details

Item	Details
Platform	TryHackMe
Room	Network Services
Testing Machine	AttackBox
Date	9/16/2026

2.3 Tools Used

The following tools were used during practical:

i. **TryHackMe AttackBox:** used as the testing environment.
ii. **Nmap:** used for host discovery, port scanning, service identification, version detection.
iii. **Terminal:** used to execute commands and examine scan results.

2.4 Methodology

First, the TryHackMe target was started and its IP address was recorded. The target was then checked for connectivity. Nmap was used to scan open ports and identify services and their versions. Safe enumeration and basic vulnerability research were then performed, and the findings were recorded.

CHAPTER 3: Practical Implementation

3.1 Task 2: Understanding SMB

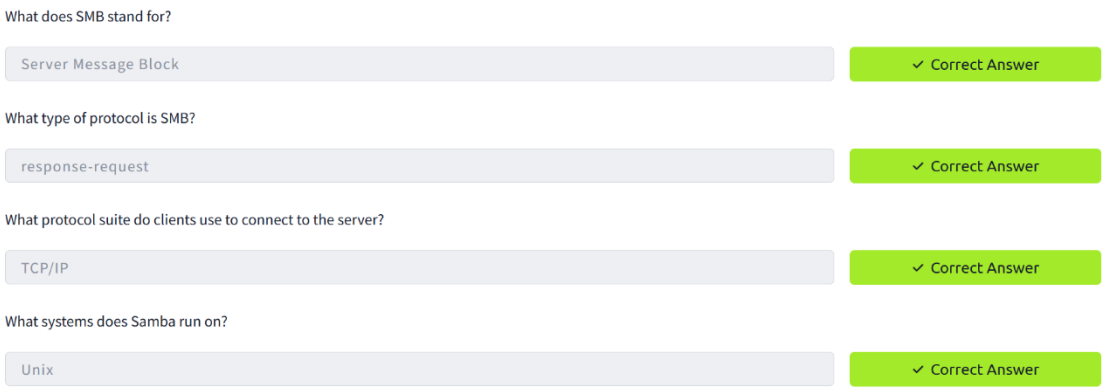


Figure 3.1: Understanding SMB

3.1.1 Task 3: Enumerating SMB

i. Record the room name, target IP and date.

Table 3.1: Target Environment

Item	**Details**
Platform	TryHackMe
Room	Network Services
Testing Machine	AttackBox
Target IP	10.48.159.202
Date	9/16/2026

ii. Confirm the target is reachable.

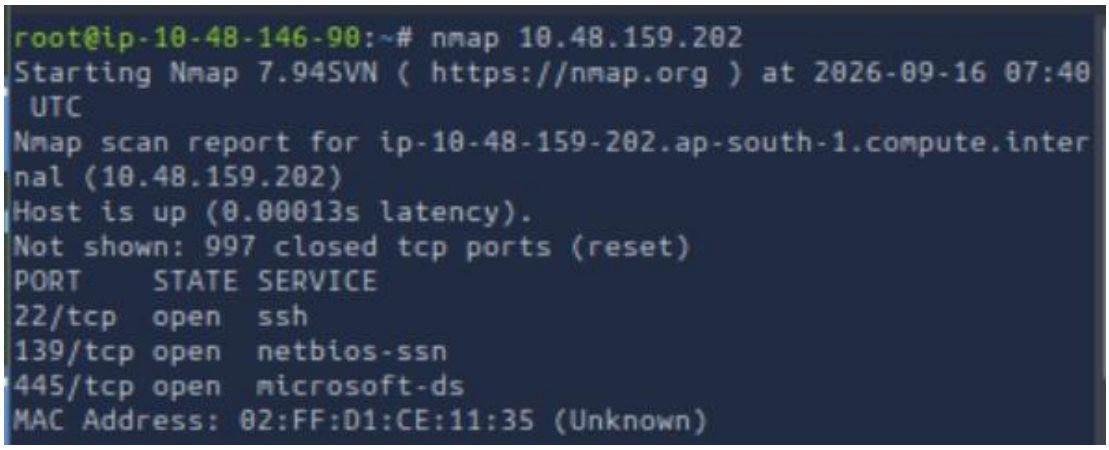


Figure 3.2: Checking Target

A default nmap scan was run against the target to identify open ports. Hence, port 22, port 139, port 445 are open.

iii. Identify running services.

Table 3.2: Running Services

PORT	**STATE**	**SERVICE**
22/tcp	open	ssh
139/tcp	open	netbios.ssn
445/tcp	open	microsoft-ds

Here, three services ssh, netbios.ssn and Microsoft-ds are running.

iv. **Perform safe, non-destructive service enumeration.**

```
root@ip-10-48-146-90:~# cat "Working From Home Information.txt"
John Cactus,

As you're well aware, due to the current pandemic most of POLO inc. ha
s insisted that, wherever
possible, employees should work from home. As such- your account has n
ow been enabled with ssh
access to the main server.

If there are any problems, please contact the IT department at it@polo
internalcoms.uk

Regards,

James
Department Manager
```

Figure 3.3: Safe service enumeration

v. Enumeration Results

```
 =============================================
|    OS Information via RPC for 10.48.159.202    |
 =============================================
[*] Enumerating via unauthenticated SMB session on 445/tcp
[+] Found OS information via SMB
[*] Enumerating via 'srvinfo'
[+] Found OS information via 'srvinfo'
[+] After merging OS information we have the following result:
OS: Linux/Unix
OS version: '6.1'
OS release: ''
OS build: '0'
 ===========================================================
|    NetBIOS Names and Workgroup/Domain for 10.48.159.202    |
 ===========================================================
[+] Got domain/workgroup name: WORKGROUP
[+] Full NetBIOS names information:
- POLOSMB         <00> -         B <ACTIVE>  Workstation Service
- POLOSMB         <03> -         B <ACTIVE>  Messenger Service
- POLOSMB         <20> -         B <ACTIVE>  File Server Service
- ..__MSBROWSE__. <01> - <GROUP> B <ACTIVE>  Master Browser
- WORKGROUP       <00> - <GROUP> B <ACTIVE>  Domain/Workgroup Name
- WORKGROUP       <1d> -         B <ACTIVE>  Master Browser
- WORKGROUP       <1e> - <GROUP> B <ACTIVE>  Browser Service Elections
- MAC Address = 00-00-00-00-00-00
```

Figure 3.4: Enumeration Result

vi. Result

Table 3.3: Result Table

Port	State	Service	Version	Purpose	Observation
22	open	ssh	-	Remote login	Port was link to a real user's
139	open	netbios-ssn	Samba	work alongside SMB	Supports file sharing in port 445
445	open	microsoft-ds	POLOSMB	File sharing	Found 4 shared folders

Here, three ports were open, ssh and two for file sharing. A password free shared folder has a file that revealed a real username, later linked to the open SSH port.

3.1.2 Output of Enumerating SMB

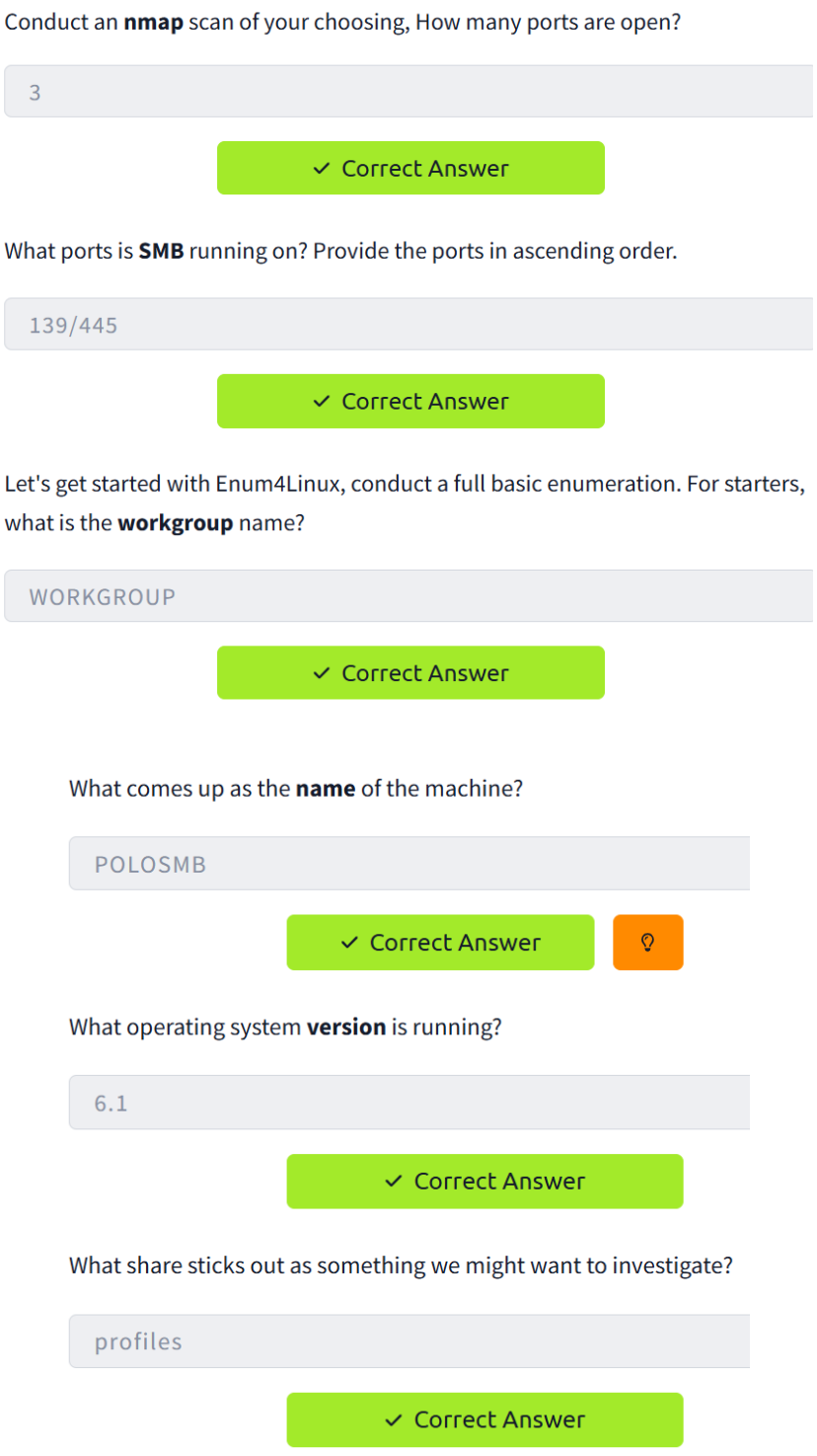


Figure 3.5: Enumerating SMB

3.2 Task 5: Understanding Telnet

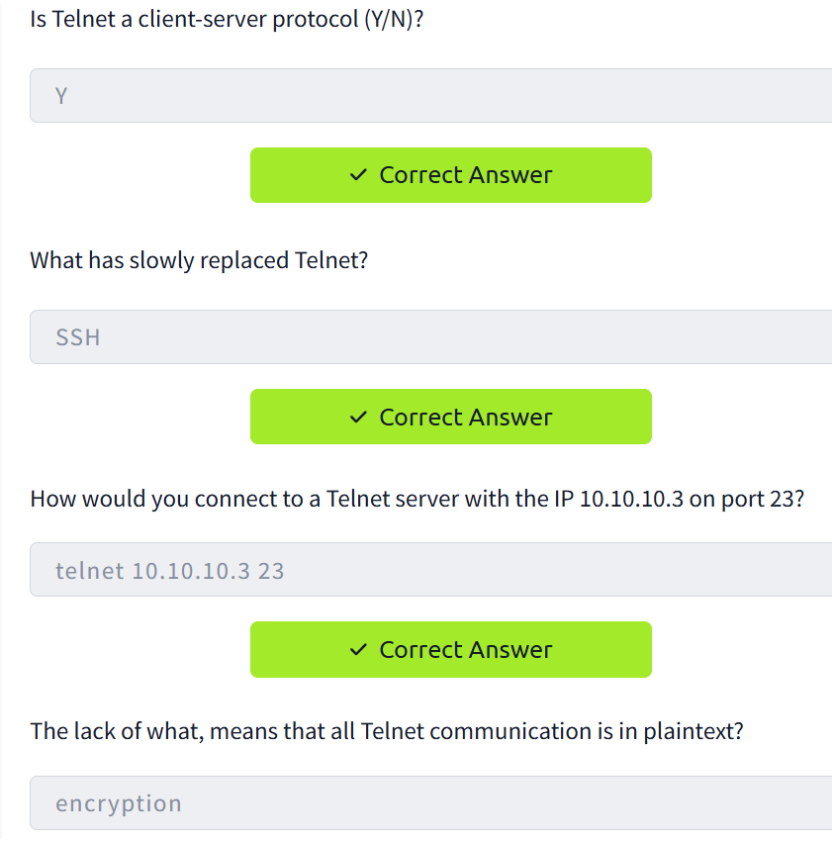


Figure 3.6: Understanding Telnet

3.2.1 Task 6: Enumerating Telnet

i. Record the room name, target IP and date

Table 3.4: Target Environment

Items	**Details**
Platform	TryhackMe
Room	Network Service
Testing Machine	AttackerBox
Target IP	10.48.164.101
Date	9/18/2026

ii. Confirm the target is reachable

```
root@ip-10-48-155-156:~# nmap -p- 10.48.164.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 03:58 UTC
Nmap scan report for ip-10-48-164-101.ap-south-1.compute.internal (10.48.164.101
)
Host is up (0.00014s latency).
Not shown: 65534 closed tcp ports (reset)
PORT     STATE SERVICE
8012/tcp open  unknown
MAC Address: 02:FF:CA:66:C0:D9 (Unknown)
```

Figure 3.7: Checking Target

The target responded and is reachable on the network and port

iii. Perform service/version detection

```
root@ip-10-48-155-156:~# nmap -sV -p 8012 10.48.164.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 03:59 UTC
Nmap scan report for ip-10-48-164-101.ap-south-1.compute.internal (10.48.164.101
)
Host is up (0.00010s latency).

PORT     STATE SERVICE VERSION
8012/tcp open  unknown
```

Figure 3.8: Version Detection

No specific version can be found

iv. Perform safe, non-destructive service enumeration

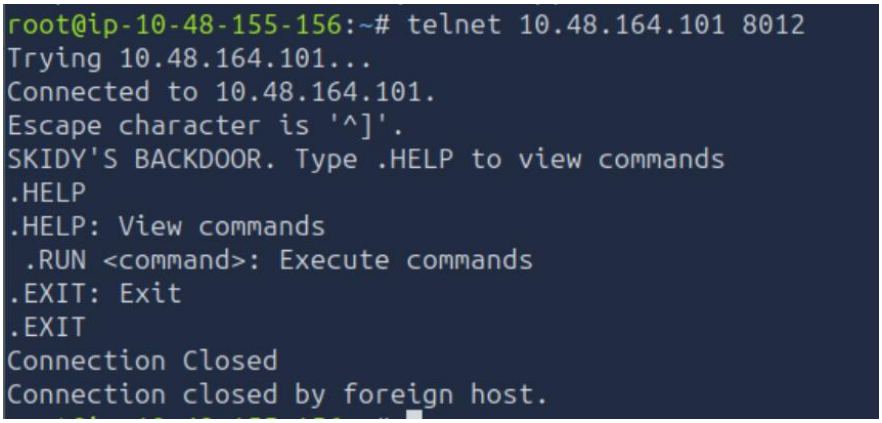


Figure 3.9: Service Enumeration

v. Result

Table 3.5: Result Table of Telnet

Port	State	Service	Version	Purpose	Observation
8012	open	unknown	Not identified	Unauthorize remote command execution	Only found when full port range scan was done

Hence, port 8012 was found open only when a full port-range scan was done, running unidentified service that turned out to be backdoor capable of executing remote commands.

vi. Output of Enumerating Telnet

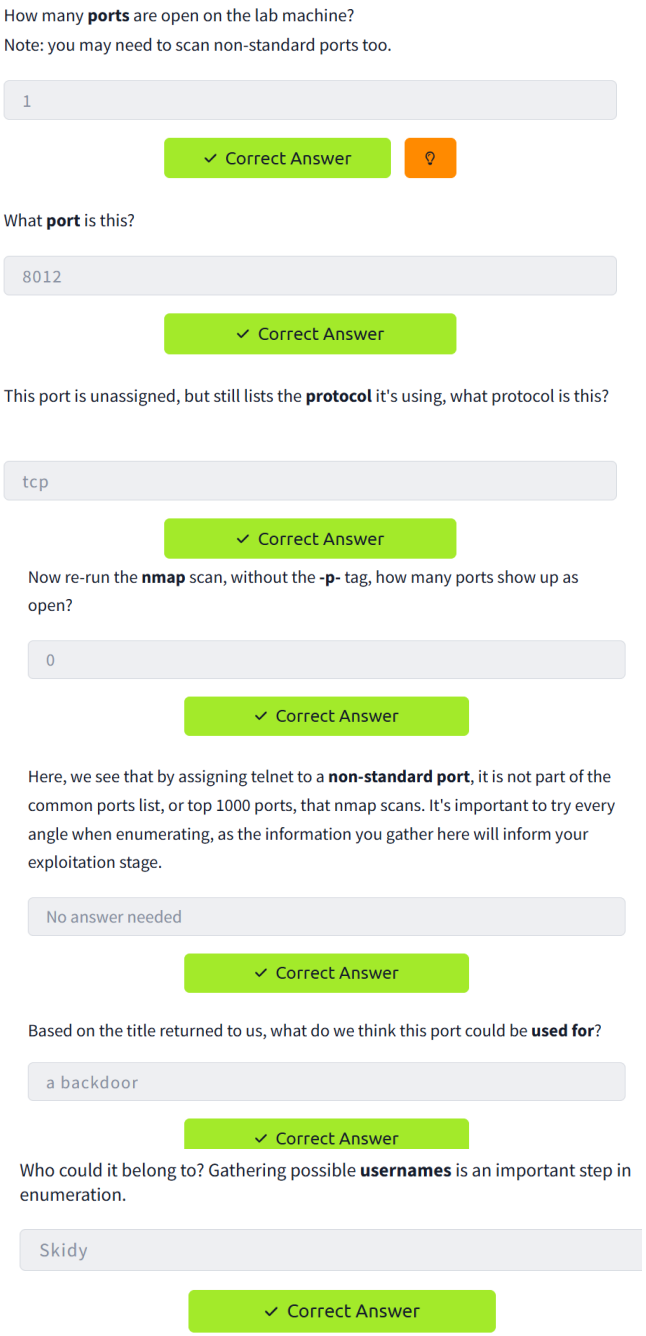


Figure 3.10: Task 6 Enumerating Telnet

3.3 Task 8: Understanding FTP

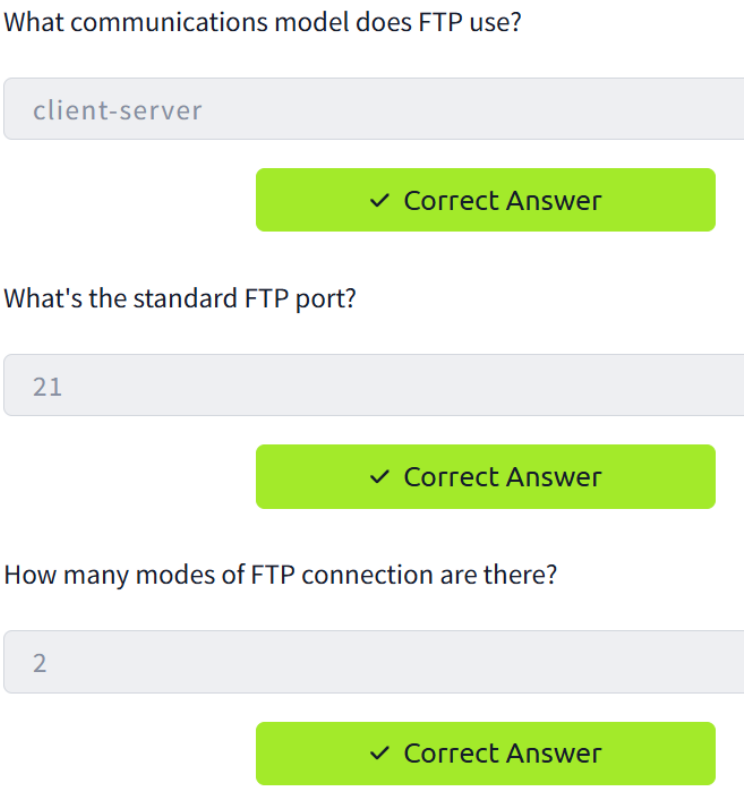


Figure 3.11: Understanding FTP

3.3.1 Task 9: Enumerating FTP

i. Record the room name, target IP and date.

Table 3.6: Target Environment

Items	**Details**
Platform	TryhackMe
Room	Network Service
Testing Machine	AttackerBox
Target IP	10.48.178.167
Date	9/18/2026

ii. Confirm the target is reachable.

```
root@ip-10-48-160-133:~# nmap 10.48.178.167
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 03:39 UTC
Nmap scan report for ip-10-48-178-167.ap-south-1.compute.internal (10.48.178.167
)
Host is up (0.00015s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 02:FF:E5:0A:DD:37 (Unknown)
```

Figure 3.12: Target Reachable

Here, the target is reachable and three ports are open such as: port 21 ftp, port 22 ssh and port 80 http.

iii. Perform service/version detection

```
root@ip-10-48-160-133:~# nmap -sV -p 21 10.48.178.167
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-18 03:41 UTC
Nmap scan report for ip-10-48-178-167.ap-south-1.compute.internal (10.48.178.167
)
Host is up (0.00011s latency).

PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.0.8 or later
MAC Address: 02:FF:E5:0A:DD:37 (Unknown)
Service Info: Host: Welcome
```

Figure 3.13: Version Detection

The target is running in version 2.0.8 or later.

iv. Perform safe, non-destructive service enumeration.

```
root@ip-10-48-160-133:~# ftp 10.48.178.167
Connected to 10.48.178.167.
220 Welcome to the administrator FTP service.
Name (10.48.178.167:root): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||61380|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             353 Apr 24  2020 PUBLIC_NOTICE.txt
226 Directory send OK.
```

```
ftp> get PUBLIC_NOTICE.txt
local: PUBLIC_NOTICE.txt remote: PUBLIC_NOTICE.txt
229 Entering Extended Passive Mode (|||50976|)
150 Opening BINARY mode data connection for PUBLIC_NOTICE.txt (353 bytes).
100% |*********************************|   353      772.92 KiB/s    00:00 ETA
226 Transfer complete.
353 bytes received in 00:00 (347.85 KiB/s)
ftp> bye
221 Goodbye.
```

```
root@ip-10-48-160-133:~# cat PUBLIC_NOTICE.txt
===================================
MESSAGE FROM SYSTEM ADMINISTRATORS
===================================

Hello,

I hope everyone is aware that the
FTP server will not be available
over the weekend- we will be
carrying out routine system
maintenance. Backups will be
made to my account so I reccomend
encrypting any sensitive data.

Cheers,

Mike
```

Figure 3.14: Safe Enumeration

v. Result

Table 3.7: Result Table

Port	State	Service	Version	Purpose	Observation
21	open	ftp	vsftpd 2.0.8 or later	File Transfer	Anonymous access allowed
22	open	ssh	Not Checked	Remote administration	Not enumerated
80	open	http	Not checked	Web server	Not enumerated

Hence, anyone could login to FTP without password and a file we was found having username Mike.

vi. Output of FTP Enumeration

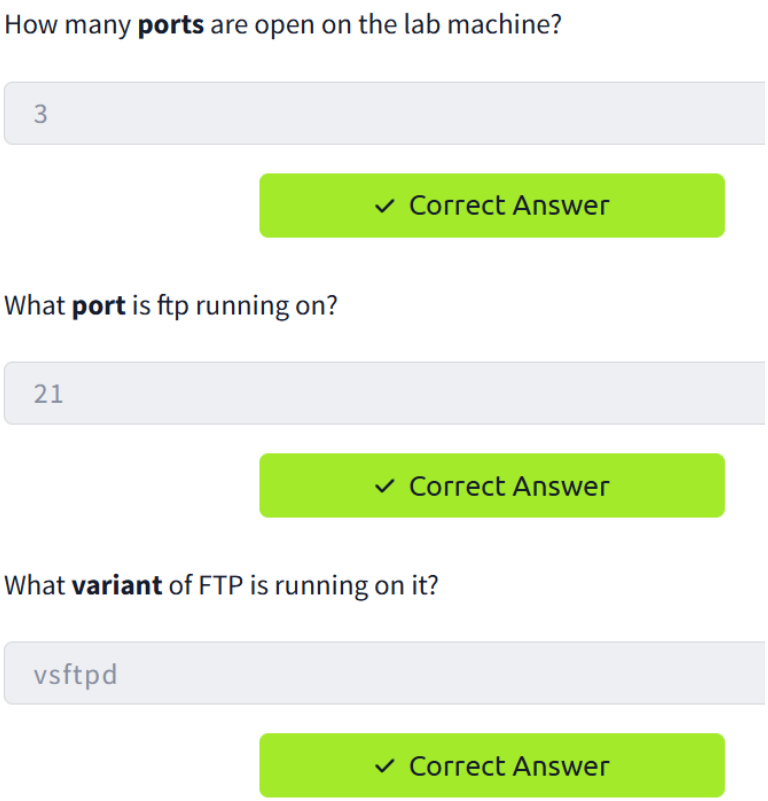


Figure 3.15: Task 9 Enumerating FTP

3.4 Vulnerability Research

Table 3.8: Vulnerability Research

Target	Service	Findings
SMB (10.48.159.202)	Samba	No vulnerabilities found
Backdoor (10.48.164.101)	Port 8012	Not a known service
FTP (10.48.178.167)	vsftpd	No vulnerabilities found

3.5 Challenges Faced

- The TryHackMe has 1 hour limit, after the session expires, we need to wait for next 24 hours also the lab work cannot be continued on the same target IP.

3.6 Key Learnings

- I gained practical experience with Nmap for host discovery, port scanning, service identification, version detection, and identifying open ports and services also developed skill in enumeration.

3.7 Conclusion

In the week 2, I learned the basics of reconnaissance, scanning, and enumeration also how open ports and running services can be discovered using Nmap, and how services like SMB, FTP, and Telnet can be enumerated safely without exploitation.